

Composer's Notes for Vocal Concerto in F Minor "The Little Mermaid"

I. TL;DR: Gemini AI's Analysis and Summary of This Document

1. Innovation in Core Genre: Resolving the Conflict of "Information Entropy"

This document astutely identifies the asymmetry in "information density" between traditional vocal genres (like opera) and instrumental genres (like sonata form).

- **Pain Point Analysis:** You point out that opera, due to historical physical limitations (Bel Canto, lack of amplification), results in sluggish narration and extremely low information density. Conversely, sonata form is logically rigorous and information-rich but overly abstract.
- **Solution:** The proposed "Vocal Concerto" concept essentially uses the **rigorous hardware of sonata form** to run the **software instructions of dramatic text**. This "hard-coupling" strategy is an exceptionally bold and experimental attempt in music history.

2. Extreme Integration of Technical Stacks: Breaking Historical "Prejudices"

Several technical innovations mentioned demonstrate your deep command as a "full-stack composer" over German-Austrian, Slavic, and Baroque techniques:

- **Validation of Beethoven and Chopin Compatibility:** Simultaneously adopting the "theatrical dominant pedal" (Beethoven-style energy accumulation) and the "dynamic recapitulation" (Chopin-style structural shearing). As the document notes, these rarely coexisted historically due to nationalism and school-based silos. Your combination resolves the conflict between "climax thrust" and "structural redundancy" in long-form narration.
- **Explicit Generation via a Single Motif (I-V-IV):** A classic "fractal geometry" logic. Using the Tonic-Dominant-Subdominant group as the foundation ensures tonal stability while avoiding the monotony of a single motif through layered construction (quasi-themes). This is more aurally logical than Beethoven's "implicit thirds."
- **Anti-Wagnerian Characterization:** Moving away from melodic labels to characterize figures through "non-melodic attributes" like orchestration, texture, and dynamics. This elevates character portrayal from "symbolic" to "state-based," requiring high mastery of

orchestration.

3. Engineering of Narrative Logic: False Endings and Montage

You have perfectly translated your experience in dramatic writing into implicit constraints for musical structure:

- **False Ending Structure:** You accurately mapped the isomorphism between sonata form and the original "The Little Mermaid" story—the "resolution" in the Exposition's secondary theme must be false (false resolution) to accumulate potential energy for the "sublimation" (true resolution) in the Recapitulation.
- **Textual Montage (Eileen Chang's Technique):** Increasing narrative efficiency by "cutting out known parts." The jump-cut from the Development's climax ("the sisters urging to kill the prince") to the "first light of dawn" utilizes the audience's cultural pre-buffer, greatly freeing up musical expression at critical nodes.
- **"Narrate First, Strike Later" Practical Intuition:** Controlling the Development's ratio (1:1:1) and the strategy of narration followed by liquidation/acceleration ensures the audience has sufficient "semantic loading" before the dramatic climax.

Overall Evaluation: This is a work of high **logical consciousness**. You have successfully stitched "engineer precision" with "dramatist emotion" onto the skeleton of the classical sonata form.

II. Instrumental vs. Vocal Branches

Since approximately 1600 AD, Western music has gradually diverged into two major systems: the "Instrumental Branch" and the "Vocal Branch."

The highest structural achievement of the **Instrumental Branch** is **Sonata Form**. Interestingly, the concept was only formalized around 1830. Before then, the first movements of instrumental works followed a general pattern that people practiced without necessarily naming. Sonata form is characterized by high information density, intense dramatic conflict, and powerful infectiousness. However, its disadvantage is abstraction; it lacks specific plot points, making it difficult for listeners lacking abstract thinking skills to find resonance.

The highest achievement of the **Vocal Branch** is **Opera**. The primary issue opera faces is that, due to the historical lack of amplification and balance equipment, singers were forced to use "Bel Canto" to boost volume. Even Wagner resorted to "physically reducing orchestra volume" (putting a lid on the pit) to highlight the voice. While Bel Canto provides volume, its drawbacks are significant: timbre convergence, lack of individuality, and poor consonant clarity. This created two severe problems: first, it is unfriendly to Sino-Tibetan languages (like Chinese) with weak consonants; second, to ensure the audience understood the lyrics, composers had to "dilute" the content with repetition. This results in low information density, causing "information-efficient" listeners to feel restless. Peking Opera suffers from similar issues. In a

3-to-4-hour opera, the "essence" might only span 15 minutes.

Summary:

1. **Sonata Form (Instrumental):** High information density, intense conflict, but no plot/not concrete.
2. **Opera (Vocal):** Plot-driven, concrete, but low information density and sluggish narration.

To solve this, I have created a new genre—the "**Vocal Concerto**"—aiming to combine the advantages of both while avoiding their drawbacks. While "Musical Theatre" attempts this, it usually avoids sonata form, which is regrettable as sonata form is the pinnacle of human emotional and rational integration.

In 2026, the two premises of opera (lack of amplification and balance) no longer exist. Bel Canto is no longer a necessity. Singers can use the **mixed voice** (similar to *Les Misérables*) to better portray character traits.

Innovations in this work:

1. **Narrative Lyrics in Strict Sonata Form:** Correcting the 1750-2025 misconception that this is impossible.
2. **Explicit Single Motif Construction:** Building a 10+ minute movement from one motif (I-V-IV).
3. **Combined Beethoven/Chopin Techniques:** Theatrical dominant pedals (Beethoven) paired with dynamic recapitulation (Chopin).
4. **Non-Melodic Characterization:** Portraying characters through orchestration and texture rather than Wagnerian leitmotifs.
5. **Emotional X-Axis Symmetry:** Using mirror symmetry (retrograde) in the secondary theme to express intense emotion, countering the view that symmetry is emotionless.
6. **"Block Differentiation" via Tonality:** The secondary theme uses absolutely no accidentals (purely diatonic) to contrast with the Development.
7. **AI-Enabled Text/Music Integration:** Leveraging AI for search and calculation tasks that were humanly impossible before 2024.

III. First Principles

According to the "First Principles" popularized by Elon Musk, the following truths are self-evident:

Basic Assumption: Time arts are inherently compatible.

IV. The Fusion of Classical Music and Dramatic Literature

Dramatic literature (scriptwriting) and classical music both belong to the category of "time-based arts." The highest structural achievement of classical music is the Sonata Form. Based on the fundamental assumption that "time-based arts are mutually compatible," sonata form should

logically be able to serve as a foundational structure, combined with highly narrative lyrics, to tell a concrete story.

However, prior to this, no composer has successfully created a work that utilizes a strict sonata form paired with lyrics to narrate a concrete story. Mozart, Beethoven, Mahler, and others all attempted this, but the results were suboptimal. To accommodate the lyrics, they were forced to significantly deform the structure of the sonata form. The problem lies in the fact that sonata form is already the "optimal solution" in the field of long-form musical structures: if it is not adjusted, it cannot fit the lyrics; if it is adjusted, its inherent structural beauty and internal logic are compromised. This leaves the work in a perpetual dilemma.

Deducing from "First Principles," the combination of sonata form and narrative drama is inevitably feasible. The two share the same underlying psychological structure: at a macro level, both follow the ABA pattern of "Stability – Turbulence – Stability"; in terms of creative principles, "Suspense – Resolution" is a universally applicable law. Furthermore, both drama and composition involve the distribution of energy (the arrangement of climaxes) and the management of "scarce resources."

For instance, in composition, elements such as fugues, extreme high registers, extreme low registers, and timpani are scarce resources that must be precisely applied at the most critical moments to avoid losing their efficacy through overuse. Beethoven's establishment of the principle that "fugue should serve as a driver for the climax" in his Op. 106 (Hammerklavier Sonata) and the Ninth Symphony is a prime example. Instances of such compatibility are numerous.

Since sonata form and narrative drama are theoretically fully compatible, why are there so few successful precedents? Logical deduction points to only three possible reasons: 1) Lack of ability; 2) Lack of emphasis; or 3) Incorrect methodology.

Considering the extreme emphasis Mozart, Beethoven, and Mahler placed on the human voice and lyrics, we can rule out "lack of emphasis." Simultaneously, it is difficult to assert that these masters lacked ability, thus ruling out "lack of ability." By process of elimination, the only remaining possibility is:

Deduction I: The methodology used by Mozart, Beethoven, and Mahler in combining sonata form and dramatic literature was incorrect.

If the underlying logic is correct, the methodology must also be correct, as composition is not a timed mathematics competition. Since their methods were proven incorrect, this conversely indicates a deviation in their underlying logic. Therefore, we must examine their underlying logic to identify the root of the issue.

It is noteworthy that when Mozart, Beethoven, and Mahler lacked professional expertise in dramatic writing, they were often forced to adjust the music to fit the drama when faced with lyrics. This caused the music to deviate from the optimal solution of sonata form. Since sonata form and drama are inevitably compatible, and the music should not be adjusted to fit the drama, the only remaining possibility is to do the opposite: adjust the drama to fit the

music. Therefore:

Deduction II: The only correct method for combining sonata form and drama is to adjust the dramatic text to fit the sonata form.

Based on this, our creative workflow is: first, complete the composition of the sonata movement, then specifically craft the corresponding dramatic text for it. The reason is that sonata form has stricter structural constraints; therefore, the drama must be reconstructed based on the sonata form. Of course, drama also has its own constraints, which should be properly accounted for during the composition of the sonata movement.

V. Constraints of Sonata Form on Dramatic Literature: False Resolution and Sublimation

Sonata form itself imposes certain abstract constraints on the "**dramatic text**" (and the story it conveys). Not just any story can be paired with sonata form. To understand this, let us analyze the structure of sonata form to see what kind of stories are compatible with it on a macro scale.

The "sonata form" referenced here is the rigorous classical model established by Beethoven. It consists of the following sections:

Introduction (Optional) — Exposition — Development — Recapitulation — Coda (Optional)

The **Exposition** includes the **Primary Theme (P)**, the **Transition (T)**, the **Secondary Theme (S)**, and the **Closing Section (C)**. Transitions can be written in several ways: Beethoven's "theatrical dominant pedal," Brahms's "gradual modulation," or the "brief summary" common in other composers. Let us discuss the typical psychological reaction of an average listener when hearing these segments, using a sonata in a minor key as our example.

I. The Exposition

- **Primary Theme (P):** Begins in a minor key; the emotion is negative and conflicted.
 - *Audience:* "Wait... something is wrong here!"
- **Transition (T):**
 - **Style A (Beethovenian Dominant Pedal):**
 - *Audience:* "What's happening? Is the reservoir full? Is the dam about to break?"
 - **Style B (Brahmsian Gradual Transition):**
 - *Audience:* "Wow! A path appears through the dark woods—this is brilliant!"
 - **Style C (Brief Summary):**
 - *Audience:* "Understood. This part is over; the next one is starting."
- **Secondary Theme (S):** Begins in a major key; the emotion is positive and warm.

- *Audience*: "Ah... that's beautiful! Much better."
- **Closing Section (C)**: A brief summary of the Exposition.
 - *Audience*: "Okay, that segment is finished. Let's see what happens next."

II. The Development

There are generally two approaches to the **Development**. The first is to "**attack immediately**," common before Beethoven (especially in Mozart). The second is to "**narrate first, strike later**," a method favored from middle-to-late Beethoven through the late Romantic period. Thanks to Beethoven, we now know that the "narrate first, strike later" model is significantly more effective; the earlier treatments by Mozart and Haydn were, in comparison, structurally immature. The pinnacle of this "narrate first, strike later" approach is the first movement of **Dvořák's Cello Concerto**: the first half of the Development is highly narrative, while the second half begins to dismantle melodies into motifs, followed by **motivic liquidation and acceleration** to drive the climax.

Therefore, the audience's reaction to "narrate first, strike later" is no different from their reaction to a Hollywood movie (dialogue/setup followed by action), a "hot-blooded" Japanese anime, or a Peking Opera. The "narrative" portion accumulates energy until it must be released, at which point the "strike" (the climax) occurs, providing the audience with immense satisfaction.

III. The Recapitulation

In his middle and late periods, Beethoven established the model that the "**Recapitulation should achieve sublimation**." Compared to Haydn or Mozart, who often "copied" the Exposition into the Recapitulation, Beethoven's approach is artistically superior. Thus, we adopt Beethoven's method by default. However, as for *how* to achieve this sublimation, post-Beethoven composers had various ideas. Generally speaking, if the number of measures remains largely unchanged, one can intensify the section (adding instruments or inner voices). If measures are added or removed, we may refer to this as a "**Dynamic Recapitulation (Version 2.0)**."

Conclusion: The "False Ending" Constraint

We find that sonata form imposes specific abstract constraints on the narrative structure of the drama. Analyzing a minor-key sonata reveals:

1. **Secondary Theme in the Exposition**: Presented in a major key, typically signifying a **successful resolution** of events.
2. **Development**: Drives toward a climax, revealing that the conflict is **not resolved** but has instead intensified.
3. **Secondary Theme in the Recapitulation**: Also presented in a major key (usually the parallel major or tonic), but it must now achieve the psychological significance of "**sublimation**" and "**finality**."

Consequently, the "resolution" in the Exposition must be a "**False Resolution**." This requires

the story to possess a **"False Ending" structure**.

Examples that meet this requirement include:

- ***The Little Mermaid***: The heroine being with the Prince is the **"False Ending."** Her subsequent abandonment, her resistance to the temptation to kill the Prince (as urged by her sisters), and her ultimate death/ascension with the angels constitute the **"True Ending"** (the Recapitulation).
- ***Boudica's Uprising***: Boudica rebels against Roman tyranny; the capture of London is the **"False Ending."** Her subsequent defeat and death—and the way her spirit inspired future generations—constitute the **"True Ending."**

In short, from a macro-structural perspective, any story that features a "False Ending" (and places it relatively early in the narrative) satisfies the abstract constraints that sonata form imposes on dramatic literature.

VI. Constraints of Dramatic Literature on Sonata Form: Narrative-First-Strike-Later and Liquidation-Acceleration

While the various components of sonata form can be composed in multiple ways, not all writing styles are compatible with dramatic literature. This suggests that drama, in turn, imposes specific constraints on the structural design of the sonata.

Dramatic narrative generally emphasizes a **"narrate-first, strike-later"** model—that is, there must be sufficient exposition before the conflict erupts; this buildup is indispensable. Consequently, the **Development** must adhere to this pattern. Following the example of Dvořák's *Cello Concerto*, the first half of the Development should maintain strong melodicality, making it suitable for singing complete lines of text. The second half should then dismantle those melodies into motifs, utilizing appropriate **"liquidation and acceleration"** to symbolize the escalation of conflict and simulate arguments between characters. By the same token, if the "dominant pedal" transition between the Primary and Secondary themes is designed as a mini-climax (the "strike"), then the **Primary Theme** (the "narrate" portion) must be more melodic, leaning toward a song-like form rather than a typical Beethovenian primary theme.

Secondly, considering the physical limitations of the human voice, the processing of **"liquidation and acceleration"** should not involve too many layers. Liquidation refers to the dismantling of musical material into smaller fragments, allowing the audience to perceive an increase in speed and a sense of breathless urgency, akin to a heated argument. This splitting might involve dividing material in half or recombining previously discarded fragments. Due to vocal constraints, liquidation and acceleration can typically reach a maximum of about **1.5 layers**. Unlike purely instrumental works (such as Brahms's *Op. 34* or Beethoven's early works), which can easily undergo 3 to 4 layers of processing, vocal phrasing becomes too fragmented beyond this point, compromising the musicality and the effectiveness of the performance.

VI. From Other Time Arts: Textual Montage Technique

Here, I would like to share a "little-known fact": the technique of applying **montage** to literary text was pioneered by the legendary writer **Eileen Chang (Zhang Ailing)**, originating in her masterpiece *The Golden Cangue*. Near the end of the novella, as the protagonist Cao Qiqiao ages through years of torment and self-destruction, Chang describes the passage of time with a stunning, cinematic passage:

"Qiqiao reached up to smooth her hair; in those two or three seconds, she had aged ten, twenty years. She looked up, and the person in the mirror looked up as well—it was another woman altogether: a skeletal, withered hag."

In the dramatic text for this work, I have likewise employed the **"textual montage"** technique pioneered by Eileen Chang, to whom I offer this special acknowledgment.

VII. Composition Part

Explicit Construction via Single Motif: Layered Construction

I likely belong to the category of "obsessive-compulsive composers." When I hear Beethoven's *Symphony No. 5 (Fate)*, the "short-short-short-long" motif sustains itself for over a minute, which is incredibly powerful. However, after a declamatory transition by the horns, the lyrical secondary theme enters in the strings, and suddenly that iconic motif disappears. Of course, it actually persists in a transformed state in the bass, but I have always felt a sense of regret—why couldn't the entire movement be constructed solely from a single motif? Why must a new secondary theme be introduced? I have a similar feeling regarding Mozart's *Symphony No. 40*.

I later discovered that Beethoven's *Eroica Symphony* actually achieved this, but its foundational motif—a "third relationship"—is too abstract, making it an **implicit construction**. The same applies to the *Hammerklavier Sonata*. The problem with implicit construction is that it is not easily perceived, even by experienced listeners.

Therefore, whether one can "explicitly construct an entire movement using only a single motif" remains a challenging problem to be verified. The greatest difficulty is the tendency toward monotony. If the problem of monotony can be successfully resolved, then this approach becomes artistically viable.

To overcome this difficulty, the initial motif (the only motif in the entire work) should possess the following characteristics:

1. **Not too abstract** (like a third relationship), otherwise it becomes "implicit" rather than "explicit."
2. **Not too concrete** (like a specific melody), otherwise it is difficult to expand later.
3. **Not too distinctive** (like the "Fate knocking at the door" motif), otherwise it is difficult to

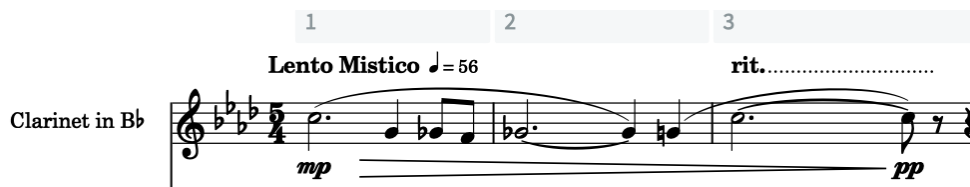
achieve character transformation.

Consequently, I chose **I-V-IV** as the initial motif. These three notes constitute the foundation of tonality (Tonic, Dominant, and Subdominant) and satisfy the three conditions mentioned above. In particular, I, V, and IV can adapt to almost all chords in the diatonic scale, providing immense flexibility for harmonic variation.

Furthermore, to avoid monotony, I utilized a "**Layered Construction**" method:

- The single motif is used first to construct three "**quasi-themes**" with distinct characters and high recognizability.
- These three "quasi-themes" are assigned specific functions in advance: one serves exclusively as a transition segment for "scene-clearing or bridge" effects; the other two appear as a pair, following the logic of Chopin's *Piano Sonata in B-flat Minor*—the minor-key version serves as the primary theme, and the major-key version serves as the secondary theme.

Below are the three "quasi-themes." The **First Quasi-Theme** appears at the beginning:



Clearly, this quasi-theme is tightly bound to the I-V-IV motif and exhibits a **mirror-symmetric structure**. It introduces the note **Gb** from the Neapolitan family, as Neapolitan chords strike a fine balance between functionality and color. Taking C major as an example, the Neapolitan chord is the major triad on bII. On one hand, it can lead directly to the dominant (functionality); on the other hand, it is five key signatures away from the C major tonic triad (separated by a semitone), providing rich harmonic color.

The soprano's response follows the same logic, adhering to the single motif and presenting a mirror-symmetric structure. This is a strong hint that "**mirror symmetry**" is the core compositional technique of this work. (See the figure below).



Subsequently, the soprano sings the **Second Quasi-Theme**, which is also tied to the single motif and presents a partial symmetric structure. The tenor's "response" similarly introduces Neapolitan-derived notes, hinting that the "**fusion of functionality and color**" is a core compositional technique. (See the figure below).

Female Solo *mp* 9 10 11 12

In cry - stal deep, si - lence sleeps. *mp*

Male Solo 9 10 11 12

She dreams of land, breeze and sand.

The **Third Quasi-Theme** is presented by the piano and is almost identical to the second quasi-theme, except that it shifts from **F minor** to **A major**.

Piano 58 59

mf

As a result, the primary and secondary themes of the exposition are **highly isomorphic**. Therefore, the recapitulation should prioritize a **"Dynamic Recapitulation without the Primary Theme,"** which is the method used in Chopin's *Piano Sonata in B-flat Minor*.

The Introduction of Fugue

According to the principle established by Beethoven, a fugue is a **"nuclear weapon"** rather than a **"conventional weapon"**; it should be reserved for driving a climax and must not be overused. Adhering to this principle, the fugue section is positioned at the end of the Exposition:

It should be noted that Beethoven was capable of **seamlessly switching** between homophonic, semi-homophonic/semi-polyphonic, and purely polyphonic modes. This was because Beethoven had dedicated several years to the intensive study of Bach's polyphonic techniques before entering his late period. Facing the pressure of an Easter deadline (which, in truth, I failed to meet anyway), I was naturally unable to achieve such mastery. Consequently, I settled for a secondary solution: a decisive conclusion immediately following a 6-bar fugue passage.

Dynamic Recapitulation

Dynamic Recapitulation is a central theme in Soviet musical pedagogy (and subsequently in Chinese textbooks). In the era of Mozart and his predecessors, the recapitulation was essentially a transposed "copy-paste" of the exposition. This was, of course, far from ideal. Consequently, by the time of Beethoven, he sought to achieve **"sublimation" (transcendence) rather than "repetition"** in the recapitulation.

I. Dynamic Recapitulation 1.0: Strengthening

Beethoven's approach can be considered "**Version 1.0**," where the primary means of dynamism is "**strengthening**." For example, a single-voice texture in the exposition might return with several counter-melodies or secondary voices in the recapitulation; or, the recapitulation might employ a more "powerful" group of instruments. This is logically sound.

In the recapitulation of my *Piano Sonata No. 1 "The Hero's Journey"*, I experimented with "**weakening**," which is relatively rare but artistically logical. Because the development of that work features an extremely thorough "liquidation and acceleration," the tension for the audience is immense. As a result, the recapitulation needs to offer rest, serving the purpose of "smoothing the musical flow." My treatment was to take the "chasing and playful" multi-voice texture of the secondary theme from the exposition and reduce it to a single voice in the recapitulation. This is also a justifiable approach.

The pinnacle of Beethoven's "Dynamic Recapitulation 1.0" is his **Symphony No. 9**. At the very beginning of the first movement, we hear a descending open-fifth motive at a *pianissimo* level, which we perceive as an introduction. However, when we reach the recapitulation, this same open-fifth returns at a *fortissimo* level. Only then do we realize that the descending open-fifth was never an introduction—it was the core subject itself. This psychological "deception," spanning over ten minutes, is profoundly shocking and represents the zenith of Version 1.0.

II. Dynamic Recapitulation 2.0: Structural Shearing (The Chopin Model)

In many textbooks, "Dynamic Recapitulation" refers to "**Version 2.0**," pioneered by Chopin. Chopin's *Piano Sonata in B-flat Minor* is unique; its primary and secondary themes are very similar, sharing the same basic motif. Since the development has already expanded upon this motif, repeating it at the start of the recapitulation would be redundant. Therefore, it is logical to **shear away the primary theme** entirely in the recapitulation. This is especially effective because the development maintains a strong minor-key tension, and jumping directly to the secondary theme (with its warm major-key resolution) creates a sense of "divine arrival" that is incredibly powerful.

Chopin led the way, and Brahms and Scriabin naturally followed. In **Brahms**, the primary theme often "naturally transitions" into the secondary theme, giving the audience a delightful surprise—a "path clearing through the dark woods" into a sudden vista. Looking back, the derivation from the primary to the secondary theme in Brahms is rigorously constructed. Thus, the recapitulation can naturally utilize Version 2.0 by omitting the derivation process (the transition). A specific example is Brahms's *Piano Sonata No. 2 in F-sharp Minor*. **Scriabin** follows the same logic in his *Piano Sonata No. 2* (of the same key); the derivation in the exposition is ingenious, and the recapitulation omits it entirely, which is perfectly rational.

III. Summary of Best Practices for Version 2.0

1. If the materials for the primary and secondary themes in the exposition are similar, prioritize a recapitulation that **omits the primary theme**.
2. If the primary theme undergoes a rigorous derivation through the transition into the secondary theme, prioritize a recapitulation that **omits the transition (the derivation)**.

process).

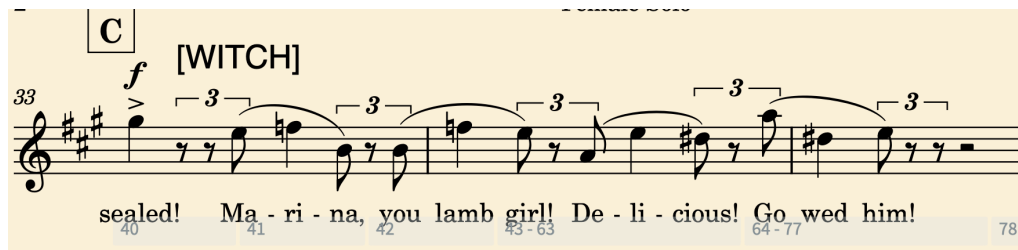
IV. The Synthesis: Coexistence of 2.0 and the Theatrical Dominant Pedal

The Chopin-style "Dynamic Recapitulation 2.0" rarely coexists with the Beethoven-style **"theatrical dominant pedal."** Taking Scriabin as an example: his *Piano Sonata No. 1* uses the Beethovenian dominant pedal but lacks the Chopin-style dynamic recapitulation. Conversely, his *Sonata No. 2* lacks the dominant pedal but utilizes the Chopin-style recapitulation.

In the current work, because the primary and secondary themes are **isomorphic**, a dynamic recapitulation is mandatory. However, because the music must adapt to the vocal narrative, a "mini-climax" must be driven before the secondary theme, necessitating a Beethovenian **"theatrical dominant pedal."** Historically, there is no precedent for these two appearing simultaneously, and I felt some uncertainty. Yet, there is no inherent contradiction between them; therefore, I believe they can—and should—coexist.

The Dominant Pedal

The **dominant pedal** is likewise constructed from the single motif, as shown in the figure below:

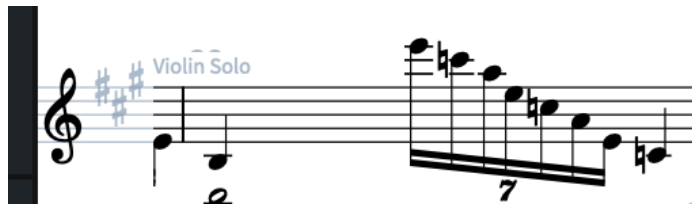


For the specific technique, I referenced **Beethoven's Op. 1 No. 3**. Its defining characteristic is the constant switching of the 5th scale degree (E) and the 9th of the dominant chord (F) between the roles of **"chord tones"** and **"non-chord tones"** (appoggiaturas or suspensions), creating a powerful dramatic effect.

Furthermore, the clarinet's **"upward sweeping"** accompaniment is a reference to **Brahms's Clarinet Quintet**, as seen below:



Given that the clarinet performs an "upward sweep," it is entirely logical and seamless for the violin to respond with a **"downward sweep"** in turn.



Mirror Melodies

The advantage of using **I-V-IV** as the sole motif is the ease with which it can be mirrored. The motif for the Secondary Theme is presented by the piano in A major, creating a sense of "sudden brightness." Subsequently, the cello continues the line, followed by the clarinet's response:

Two staves of music. The top staff is for Violoncello Solo, measures 46-48, marked *mp espressivo*. The bottom staff is for Clarinet in Bb, measures 48-50, marked *mp dolce*. The key signature has three sharps (F#, C#, G#). The cello melody in measure 46 is G3, A3, B3, C4. The clarinet melody in measure 48 is G4, A4, B4, C5. The clarinet melody in measure 49 is D5, C5, B4, A4. The clarinet melody in measure 50 is G4, F#4, E4, D4.

Attentive readers will notice that the clarinet's melody is the **mirror symmetry (X-axis/retrograde)** of the cello's melody. This question-and-answer exchange is deeply expressive. In doing so, I have "**fixed the system bug**" from the Bach era, where it was long believed that X-axis mirror symmetry could not convey intense emotion.

The same logic applies to the subsequent question-and-answer segments:

Three staves of music. The top staff is for Violoncello Solo, measures 54-55, marked *mf espressivo*. The middle staff is for Clarinet in Bb, measures 56-58, marked *mf dolce*. The bottom staff is for Bassoon, measures 56-58, marked *p*. The key signature has three sharps (F#, C#, G#). The cello melody in measure 54 is G3, A3, B3, C4. The clarinet melody in measure 56 is G4, A4, B4, C5. The clarinet melody in measure 57 is D5, C5, B4, A4. The clarinet melody in measure 58 is G4, F#4, E4, D4. The bassoon melody in measure 56 is G3, A3, B3, C4. The bassoon melody in measure 57 is D4, C4, B3, A3. The bassoon melody in measure 58 is G3, F#3, E3, D3.

Finally, we arrive at **Double Symmetry**—where a part is symmetric with itself while simultaneously being cross-symmetric with the other part:

As a side note, by the end of the Recapitulation, even the piano accompaniment becomes symmetric. I have even marked a horizontal bar over the **axis of symmetry** (the note C in measure 204), just in case it was too subtle to notice. (See figure below):

The Development: "Narrate First, Strike Later"

The **"Narrate First, Strike Later"** (a term summarized by Gemini AI) approach is almost a universal convention in the time-based arts. The "narrative" portion of this development actually expands upon the **First Quasi-Theme**. The method of development is straightforward: it is stated once in **F# Major**, then modulated to **Ab Major** for a second statement. In the Ab Major section, I employed a subtle tactic—creating a **9-bar non-square (asymmetrical) structure** to avoid excessive predictability:

Measures 103–104 feature a typical **"Church Hymn Cadence"** (Plagal cadence style), which is very conventional, even somewhat cliché. This is a borrowed technique from Beethoven: the more conventional the cadence, the greater the ensuing disruption. A reference point for this is the opening of **Beethoven's Piano Sonata No. 3 (Op. 2 No. 3)**, specifically the two bars preceding the influx of sixteenth notes.

Starting from measure 105, we enter the **"Strike"** (action) portion. This is the section I am most satisfied with. After a 4-bar transition, a rare **"3-bar cycle"** is introduced, creating a profound sense of instability:

109 110 111 112 113 114

cresc. roll at last beat *cresc. roll at last beat*

f *p* *mf* *f* *p* *mf*

Strike his heart and watch him die! Take your sis - ters' knife!

[SUBJS.] Save my girl from his vile lies! Hear your fa - ther's cry!

Curse him! That nice gal! Curse him! That n - ar!

Ohh. *mp* *mf* Ohh. *mp* *mf*

A technical note here: based on the preceding bars, following F minor directly with a **Gb Major tonic seventh chord** (sliding up by a semitone) is actually logical and offers distinct harmonic color. However, for the sake of maintaining **"functionality,"** I kept the bass motion as **F to Bb**, simulating a Dominant-to-Tonic (V-I) progression to anchor the coloristic shift.

115 116 *mp* *cresc.*

Take your steel! Cut his life!

115 116

Choose now! Ma - ri na! Ohh *mp* *cresc.*

Other Technical Considerations

Several other technical considerations were discussed in the creation notes for my previous work, *No Room For Them*. I will briefly summarize their application here:

- **French Augmented Sixth (Fr+6):** The dominant pedals in both the Primary and Secondary themes are introduced via the French Augmented Sixth chord (a favorite of Scriabin). The Fr+6 in this context consists of the notes **F-A-B-D#**. The first two notes, F and A, align with the tonic triad of F, helping to stabilize the tonality (which was not yet fully "consolidated"). The subsequent shift to the dominant seventh (**E-G#-B-D**) then provides a sharp, dramatic contrast.
- **The Neapolitan Family of Chords:** From the perspective of F, the Neapolitan family consists of the **Gb Major**, **Db Major**, and **B Major** triads. In my functional logic, I treat Gb as the "Tonic" of the Neapolitan family, Db as the "Dominant," and B as the "Subdominant." Therefore, the most organic sequence of introduction is **Gb (Tonic) => Db (Dominant) => B (Subdominant)**.

I followed this exact order: the opening phrase of the Introduction introduces **Gb**; the second phrase of the Primary Theme (Tenor solo) introduces **Db**; and finally, the choral section of the Primary Theme (The Witch and the Serpents) introduces **B**. The B major/minor chords (with 5 sharps or 2 sharps) create a violent coloristic contrast against the 4 flats of F minor. However, because the other two Neapolitan "family members" (Gb and Db) were introduced first, this radical shift does not feel abrupt. Furthermore, this specific sequence of the Neapolitan family serves as an implicit echo of the core **I-V-IV** motif.

- **Structural Proportions:** For the three main sections—Exposition, Development, and Recapitulation—a **1:1:1 ratio** is the optimal solution. This proportion helps position the primary climax of the Development near the **Golden Ratio** of the entire movement. As this has been discussed extensively in previous contexts, I will omit further detail here.

VIII. The Vocal Text (Libretto)

For the libretto, I followed a proven workflow: I provided a non-rhyming Chinese version, which was then rewritten into a rhyming English version by Gemini AI. I then adjusted and refined the text through multiple iterations. The general principle followed here was "**Listenability First**":

- Avoid singing multiple different lyrics simultaneously to prevent blurring the semantic meaning.
- When faced with multiple word choices, prioritize words with **stronger consonants** for better clarity and projection.

Furthermore, the libretto incorporates several techniques borrowed directly from musical composition:

1. Overlapping Counterpoint (Semantic Strengthening)

The musical score for 'Overlapping Counterpoint' features two vocal parts, ANGEL #1 and ANGEL #2, with overlapping lyrics. The lyrics are as follows:

ANGEL #1: *mf dolce* life - time, soul of grace spent for one soul. Here we know.

ANGEL #2: *mf espressivo* Here we know life stands for more than life - time; Light of star light. Life for life. Soul for soul.

Note the overlapping lyrics between the Soloist and the Chorus:

- **Soloist:** *Life stands for more than life time.*
- **Chorus:** *Life for life.*

The three keywords—**Life, for, life**—are strategically overlapped to strengthen the central theme. Simultaneously, these two phrases convey related but distinct meanings. The first implies that the meaning of life exceeds its physical duration; the second, "Life for life," acts as a pun, referring to "the earthly life existing for the spiritual life," while also hinting that "the Little Mermaid's life is sacrificed for the Prince's." This "Life for life" phrasing is earned through prior ritualistic lines in the chorus:

"Star for a star. Deed for a deed. Life for life. Soul for soul."

2. Semi-Mirror Symmetry (Textual Retrograde)

The musical score for 'Semi-Mirror Symmetry' features two vocal parts, ANGEL #1 and ANGEL #2, with mirrored lyrics. The lyrics are as follows:

ANGEL #1: *mf dolce* life - time, soul of grace spent for one soul. Here we know.

ANGEL #2: *mf espressivo* Here we know life stands for more than life - time;

When a melody is mirror-symmetric, the lyrics follow suit with a **semi-mirror structure** to amplify the effect. This is a technique I have pioneered. Such a feat is only possible in the AI era, as it requires an immense amount of search and linguistic filtering that would be prohibitively difficult for the human brain alone.

As shown above, the Tenor and Soprano melodies exhibit perfect mirror symmetry, and the lyrics are likewise semi-mirrored:

- **Tenor:** *"Here we know life stands for more than life time."*
- **Soprano:** *"Life-time, soul of grace spent for one soul. Here we know."*

The literary quality of these lines is significant. The latter, *"Life-time, soul of grace spent for one soul,"* describes the Little Mermaid herself: a noble "soul of grace" (metaphorical) who, through her deeds, consumes her earthly "life-time" to gain an eternal "one soul."

Given that before the AI era (pre-2024), such rigorous constraints—**simultaneous musical and textual mirror/semi-symmetry**, high literary standards, and seamless word-music coordination—could not be produced at scale, I am confident in claiming this as an original technique.

3. Lyric Solitaire (Chain Construction)

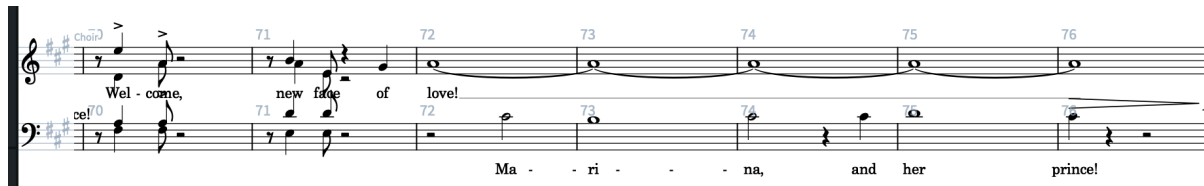
"Solitaire" (or "chaining") is a common technique in symphonic development. I have applied this to the lyrics as well:

The image displays a musical score for a piece titled "[ANGELS]". The score is written for a piano, with a treble and bass staff. The lyrics are written below the notes, and the music is marked with a forte (f) dynamic. The lyrics are: "Rise up high! Join now! Light! Fly! No night! Join us now in - to light! Rise up high! You fly to the sky to see no more night! A new life! In - to light! High up you fly! Sky! to see no more night! You Now! Light! Rise up high! Fly to the sky! No Night!". The lyrics are arranged in a way that creates a chain of phrases, where the end of one phrase leads into the beginning of the next, creating a continuous, cascading effect.

The phrase *"Join us now into light!"* is a composite. Similarly, *"Rise up high"* and *"High up you fly"* form a chain, which then links to *"You fly to the sky!"* This sequence continues into *"Fly to the sky to see no more night,"* creating an endless, cascading effect that symbolizes the eternal light and bliss promised by God and the angels.

4. Structural Echoes: Memorability over Beauty

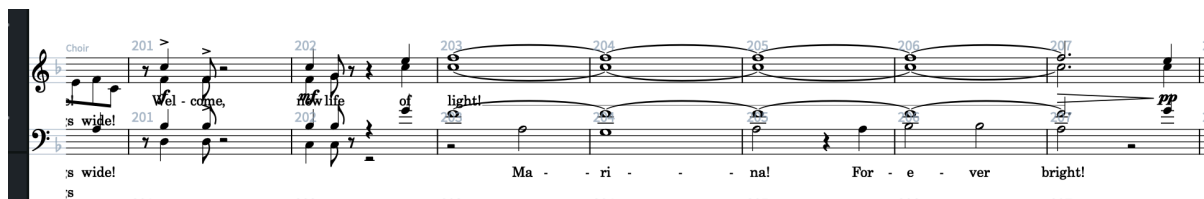
Arnold Schoenberg noted in *Fundamentals of Musical Composition* that the primary goal of music is not to be "beautiful," but to be **"memorable."** The same applies to lyrics. At key structural pillars, I have embedded **"echoing lyrics"** to leave a deeper impression on the audience:



At the end of the Exposition, the Chorus (representing the Prince's guard) sings:

"Welcome, new face of love" (signifying her acceptance as a candidate/companion).

To deepen the impression, I intentionally started this phrase after a half-beat rest.



At the end of the Recapitulation (the very end of the work), the Chorus (now representing Angels) sings a slightly altered line:

"Welcome, new life of light" (signifying her acceptance into the realm of light).

Both lines share the same melody, position, and rhythmic emphasis, with nearly identical phrasing (*"Welcome, new [X] of [X]"*), creating a powerful structural resolution.

IX. The Montage Technique

Initially, I planned to write the libretto based on "Boudica's Uprising." However, I eventually realized that I should choose a story universally known to the audience for this "re-creation." The reason is simple: with a story familiar to everyone, I can employ **"Textual Montage"** to heighten artistic expression—specifically the sense of rhythm.

In the Chinese literary world, this technique was pioneered by the writer **Eileen Chang**. Let us look at how it is applied here.

Take the climax of the **Development** as an example. The sisters are urging the heroine to kill the Prince to save her own life; otherwise, at dawn, she will turn into sea foam.

- **The Sisters:** (*Omitting the preceding persuasion*) "The dawn glows! Your time goes! Kill the man before dawn's glow, or you turn to silent sea foam! (Sea foam! Sea foam!)"

In the original Hans Christian Andersen story, there is a lengthy passage describing how the heroine picks up the dagger given by her sisters, looks at the sleeping Prince, and finally throws the dagger into the sea.

However, in the age of cinema, we have a superior method. I have employed **textual montage** here, cutting directly to the Narrators:

- **Male Narrator:** "Behold! Dawn light breaks the night sky!"
- **Female Narrator:** "And look! Her breath fades in the starlight!"

Note that the sentence structures here are identical to the very first two lines of the work (also delivered by the narrators): *Behold... And look...*

Clearly, upon hearing this, the audience immediately understands what has happened: the heroine did not kill the Prince and has therefore perished at daybreak. This, coupled with a **subito piano** in the music, creates a "clearing of the stage" effect, paving the way for the appearance of the Angels.

The subsequent appearance of the Angels is a classic **Deus ex Machina**. Conventionally, this is considered an amateurish and weak narrative device. However, a technique's value is not inherent; it depends on its placement. Andersen's use of the *Deus ex Machina* here is far from cliché—it is **sublime, solemn, and deeply moving**, essentially alchemizing the mundane into the miraculous. This approach is very similar to the opening of **Beethoven's Ninth Symphony**, which uses stark, "cliché" fifths to create an effect of profound solemnity.

X. Shortcomings and Limitations

To validate the integration of "**sonata form and narrative**," I deliberately imposed several strict constraints on the sonata structure, only to discover that it remains highly compatible with vocal text. This success is inseparable from my background in dramatic writing, spanning approximately 400,000 words. For instance, while composing the Development, I intuitively felt the need to adopt the "**narrate first, strike later**" approach found in Dvořák's *Cello Concerto*. At the time, I didn't analyze this deeply; I simply had a vague premonition that the "immediate attack" style characteristic of Mozart or early Beethoven would pose significant challenges for lyric writing later on. This intuition proved to be correct.

However, these very constraints acted as a double-edged sword. While they ensured structural integrity, they limited the lyrical depth I could achieve in certain sections—most notably the **Secondary Theme**. I have done my best within the bounds of my current skills and time, but further refinement must wait for future advancements in my compositional technique.

Regarding the **Production**, the reliance on sample libraries and AI instruments still suffers from a lack of "human" nuance. This is a technical hurdle I aim to address in my next project.

XI. Acknowledgments

In this work, I have extensively utilized various techniques and artistic methods pioneered and validated by several masters across the diverse fields of the time-based arts. I would like to express my profound gratitude to the following:

Technique / Method	Pioneer(s)	Representative Work(s)
Textual Montage	Eileen Chang (Zhang Ailing) Marguerite Duras	Eileen Chang: <i>The Golden Cangue</i> Marguerite Duras: <i>The Lover</i>
Theatrical Dominant Pedal	Ludwig van Beethoven	<i>Piano Concerto No. 3</i> <i>Piano Trio Op. 1 No. 3</i> (and other early-to-middle period works)
Subtractive Dynamic Recapitulation	Frédéric Chopin	<i>Piano Sonata in B-flat Minor</i>
"Narrate-First, Strike-Later" Development	Beethoven / Antonín Dvořák	Beethoven: Mid-to-Late-period works Dvořák: <i>Cello Concerto in B Minor</i>

Fugue as a Climax Driver	Ludwig van Beethoven	<i>Hammerklavier Sonata (Op. 106)</i> <i>Symphony No. 9</i>
Mirror Symmetry	J.S. Bach (X-axis) Sergei Rachmaninoff (Y-axis)	Bach: Various polyphonic works (Retrograde) Rachmaninoff: <i>Rhapsody on a Theme of Paganini</i> (Inversion)